Alan Binu, Junior Engineer - AI

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Summary

Results-driven technology professional with experience in building intelligent systems, automation solutions, and real-time applications. Skilled at transforming complex problems into scalable, efficient, and user-focused solutions through a strong combination of technical expertise, structured problem-solving, and innovative thinking. Passionate about AI, automation, and emerging technologies, with a continuous-learning mindset and a strong focus on delivering high-quality, measurable outcomes. Proven ability to take ideas from concept to implementation while maintaining performance, reliability, and real-world impact.

Work Experience

Apr 2025 - Present | Junior Engineer – Artificial Intelligence, Hind Rectifiers Limited | Mumbai

- Designed and delivered end-to-end **AI, Computer Vision, Robotics, and automation solutions**, from prototyping to deployment and real-world validation.
- Built **Generative AI, RAG, and Agentic AI systems** for enterprise knowledge retrieval, multi-agent orchestration, and workflow automation.
- Developed **real-time Computer Vision systems** for object detection, tracking, counting, face recognition, liveness detection, and visual inspection.
- Developed **Humanoid Robotics and Vision-Language-Action systems**, including multi-camera data collection, manipulation workflows, robot state/action recording, and NVIDIA GR00T-based development.
- Engineered a **Robot Vision and Teleoperation platform** integrating live camera streaming, AI perception, ROS2 control, browser-based teleoperation, FastAPI, and WebSockets.
- Built **ML/MLOps pipelines** covering training, evaluation, experiment tracking, model versioning, deployment, monitoring, drift detection, and retraining.

Technical Skills

AI & GenAI: LLMs, RAG, Agentic AI, LangGraph, LangChain, CrewAI, Prompt Engineering, Embeddings, Vector Search

Machine Learning: PyTorch, Scikit-learn, XGBoost, CNNs, Transformers, Fine-Tuning, Feature Engineering, Model Evaluation

Computer Vision: OpenCV, YOLO, Object Detection, Tracking, Segmentation, Face Recognition, Real-Time Inference

MLOps: MLflow, DVC, Experiment Tracking, Model Registry, Docker, CI/CD, Model Monitoring

Robotics & VLA: ROS2, NVIDIA GR00T N1.7, VLA Models, Robot Perception, Teleoperation, Nav2, SLAM

Development: Python, JavaScript, FastAPI, React, PostgreSQL, pgvector, Redis, WebSockets, Git,

Education

2025 — B.Tech in Computer Science and Engineering, MIT-WPU
CGPA-8.27

2021 — Upadhye College Of Science
Percentage-83%

2019 — Nashik Cambridge School
Percentage-86.20%